

American International University–Bangladesh (AIUB)

Department of Computer Science, Faculty of Science and Technology

Mid-Term Examination Suggestions

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Answer all questions.

Total Marks: 30

(20 × 1.5 = 30)

1. Which of the following is considered the FIRST activity in a typical Software Development Life Cycle (SDLC)?

- A. System Testing
- B. Requirement Analysis
- C. Maintenance
- D. Deployment

Answer: B

2. Which software characteristic makes software engineering significantly different from traditional engineering disciplines?

- A. Software is intangible.
- B. Software is constructed using steel.
- C. Software deteriorates because of corrosion.
- D. Software requires mechanical repair.

Answer: A

3. Which software process model is most suitable for projects involving high technical risks?

- A. Waterfall Model
- B. Spiral Model
- C. RAD Model
- D. Incremental Model

Answer: B

4. In the Waterfall Model, which phase immediately follows System Design?

- A. Maintenance
- B. Requirement Analysis
- C. Implementation
- D. Deployment

Answer: C

5. Which of the following documents primarily describes WHAT the software should accomplish rather than HOW it will be implemented?

- A. Source Code
- B. Software Requirements Specification (SRS)
- C. Test Report
- D. User Manual

Answer: B

6. Which stakeholder is primarily responsible for defining business goals during Requirements Engineering?

- A. Database Administrator
- B. Customer or Client
- C. Software Tester
- D. Network Engineer

Answer: B

7. Which Agile principle encourages delivering software at regular, short intervals?

- A. Comprehensive documentation
- B. Continuous delivery of valuable software
- C. Contract negotiation
- D. Sequential development

Answer: B

8. Which Agile artifact is commonly used to visualize the remaining work in a Sprint?

- A. Use Case Diagram
- B. Burndown Chart
- C. Context Diagram
- D. ER Diagram

Answer: B

9. Which XP practice improves software quality by simplifying existing code without changing its functionality?

- A. Refactoring
- B. Pair Programming
- C. Planning Game
- D. Small Releases

Answer: A

10. Which XP practice requires developers to write automated tests before writing production code?

- A. Continuous Integration
- B. Coding Standard
- C. Test-Driven Development
- D. Collective Ownership

Answer: C

11. Which Scrum artifact represents the completed work that satisfies the Definition of Done?

- A. Product Increment
- B. Sprint Goal
- C. Product Vision
- D. Product Roadmap

Answer: A

12. What is the primary objective of a Sprint Retrospective?

- A. Release the software.

- B. Demonstrate the product.
- C. Improve the team's development process.
- D. Estimate project cost.

Answer: C

13. Which DSDM principle emphasizes that every team member should actively participate throughout the project?

- A. Collaborate
- B. Build Incrementally
- C. Focus on Business Need
- D. Deliver on Time

Answer: A

14. Which DSDM phase confirms that the completed system satisfies business objectives before deployment?

- A. Business Study
- B. Functional Model Iteration
- C. Deployment
- D. Post-Project

Answer: C

15. Which FDD activity creates an object model that represents the overall problem domain?

- A. Plan by Feature
- B. Develop Overall Model
- C. Design by Feature
- D. Build by Feature

Answer: B

16. Which FDD role provides expert knowledge about the application domain?

- A. Toolsmith
- B. Chief Programmer
- C. Domain Expert
- D. Build Engineer

Answer: C

17. Which software engineering term refers to the ability of software to be modified efficiently after delivery?

- A. Reliability
- B. Maintainability
- C. Portability
- D. Security

Answer: B

18. Which testing level verifies the interaction among integrated software modules?

- A. Unit Testing
- B. Integration Testing
- C. Acceptance Testing

D. Regression Testing

Answer: B

19. Which document is typically prepared before coding begins to describe how software quality will be evaluated?

- A. Quality Assurance Plan
- B. User Guide
- C. Release Notes
- D. Installation Manual

Answer: A

20. Which software engineering activity ensures that modifications do not introduce unexpected defects into existing functionality?

- A. Smoke Testing
- B. Regression Testing
- C. Alpha Testing
- D. Beta Testing

Answer: B